

Multi-agent reinforcement learning framework for real-time scheduling of pump and valve in water distribution networks

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ABSTRACT

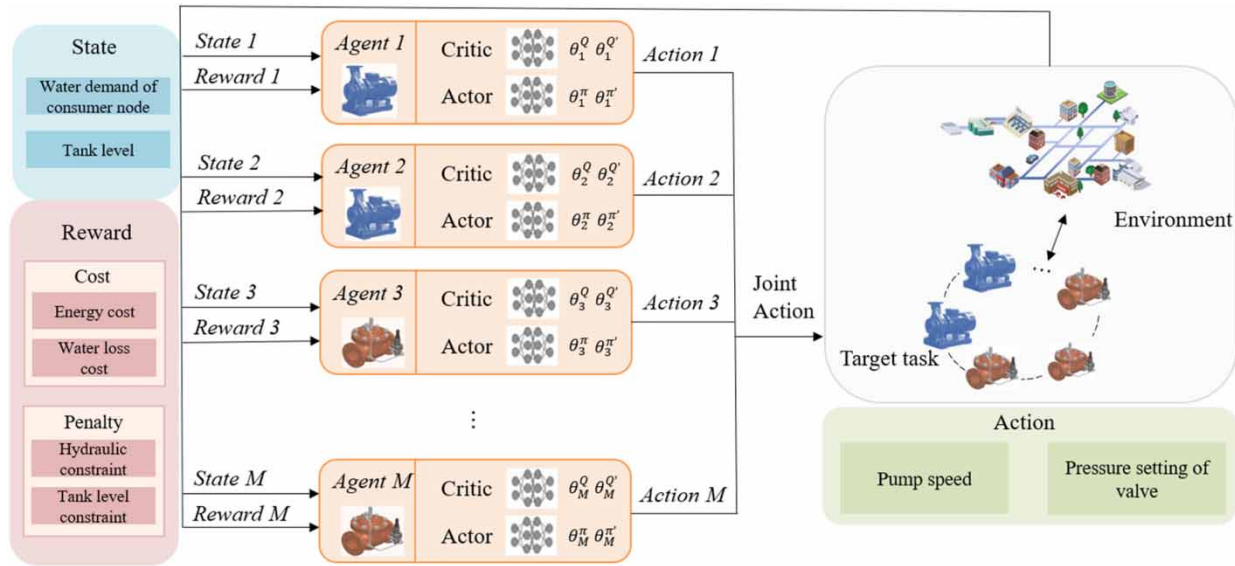
With energy and water resources shortages, the energy and water resources managements of water distribution networks (WDNs) have become increasingly important. However, achieving real-time scheduling of pump and valve in dynamic environments remains challenging. Thus, this study proposes a multi-agent reinforcement learning scheduling framework to address the uncertainty of water demand in WDNs. First, we constructed a WDN environment and modelled the scheduling problem as a Markov decision process. Second, a multi-agent deep deterministic policy gradient (MADDPG) method was used to determine the strategy of the fully cooperative multi-agent task. Moreover, the impacts of energy and water loss costs on the scheduling strategy were explored. Finally, the results were compared with those of a genetic algorithm (GA), particle swarm optimisation (PSO), and differential evolution (DE) to verify the performance and robustness of the proposed model. The results show that water loss dominates the scheduling process, and the scheduling solutions for minimising water loss and energy costs are mainly affected by the demand pattern of consumers rather than the energy tariff. The proposed MADDPG model outperforms the GA, PSO, and DE models, achieving a significantly faster solution, which is advantageous for practical applications.

Key words: energy cost, multi-agent reinforcement learning, pump and valve, real-time schedule, water distribution network, water loss cost

HIGHLIGHTS

- A multi-agent reinforcement learning framework was constructed for real-time scheduling of pump and valve.
- The scheduling problem of pump and valve was transformed into a Markov decision process.
- Scheduling solutions for minimising water loss and energy costs are affected by the demand pattern of consumers rather than the energy tariff.
- The MADDPG model can obtain a real-time solution that is superior to the GA, PSO, and DE in a relatively short time.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Water distribution networks (WDNs) are complex and difficult to manage due to several factors, including the growing populations, the dynamic nature of WDNs owing to consumer demand uncertainties, and the persistent problem of water losses within the networks (Adedeji *et al.* 2022). Inefficient operation of WDNs can lead to not only wasted energy consumption of pumps but also excessive pressure, which further contributes to water losses through leaks (Gao *et al.* 2014). The energy consumption of pumps represents 90–95% of the total energy used in WDNs (Abdelsalam & Gabbar 2021). Even a small fraction of waste can result in substantial cost increases. In addition to the wasted energy consumption of pumps, increased water losses also directly impact the economic benefits of water utilities. Water losses encompass not only financial losses from valuable water resources but also elevated maintenance costs for repairs, and additional energy required to maintain desired service levels. Considering the practical challenges of economic operation, it is crucial to enhance the energy efficiency of pumps and the water resource utilisation efficiency of WDNs.

Optimal operation of the pump and valve can reduce energy consumption and water loss. However, the cost of background leakage conflicts with the energy cost of pumps (Giustolisi *et al.* 2013). During night-time, more water is pumped to supplement tank storage, taking advantage of low energy tariffs and demand levels. However, pumping during the night causes a significant increase in water loss costs (Dai & Viet 2021). Thus, minimising leakage and energy costs is a complex problem, considering the trade-offs between objectives.

Additional computation for the leakage representation within the hydraulic model is required to quantitatively analyse water losses. The leakage demand significantly varies with changes in pressure. Giustolisi *et al.* (2008) computed the background leakage at the pipe level based on the average pressure of the pipe. This method was simulated using a pressure-driven strategy instead of fixing the demands a priori. Tolson & Khedr (2014) simulated the background leakages by adding the EPANET emitter. The average pressure of a pipe is approximated by the pressure of the end nodes. Giustolisi *et al.* (2016) pointed out that, except for the case where the leakage exponent is 1, this simplification will cause errors in evaluating the leakages. Eck *et al.* (2014) developed a technique using the emitter element in EPANET and iteratively calculate the coefficient of leakage until convergence; however, the computational load increased.

Due to the competing objectives, the need for additional computation for leakage model, and the requirement for effective cooperation between pumps and valves, the joint scheduling of pumps and valves in WDNs is a challenging issue. Gao *et al.* (2014) presented an algorithm based on the optimal control of separated valves and pumps, adding the water losses cost onto the energy cost. Shao *et al.* (2019) took the pressure-reducing valves (PRVs) and variable-speed pumps as decision variables and used a genetic algorithm (GA) to minimise the cost contributed by leakage and energy consumption under multiple working conditions. Brentan *et al.* (2018) proposed a hybrid and near-real-time optimisation algorithm of pumps and valves based

on the particle swarm optimisation (PSO) model which was linked with the real-time water demand forecasting model. The hybrid algorithm can reach the optimal solution for each time step in approximately 15 min. However, existing studies primarily focus on metaheuristic models, which have limitations in incorporating uncertainty parameters such as water demand and solving genuine real-time operational problems.

This study proposes a multi-agent framework to achieve real-time scheduling of pump and valve in dynamic environments. A multi-agent deep deterministic policy gradient (MADDPG) method is used to deal with the cooperation of pumps and valves and the uncertainty of water demand. There are huge amounts of uncertainties behind the pump and valve scheduling problems, such as the fluctuation of the water demand, while MADDPG can adapt to the uncertain environment and approach the optimal solution with minimal computation time. Considering the complexity of the objectives, different optimisation tasks are established to explore the effects of energy and water loss costs on the optimal strategy. Then, we compare the proposed model with metaheuristic models, including GA, PSO, and differential evolution (DE). The results show that water loss dominates the scheduling process; hence, the optimal solution prefers pumping following the demand pattern. By converting a large number of calculations of the training process offline, the proposed method achieves superior solutions compared to metaheuristic models, while significantly reducing the computational time required in the application. This study provides a new paradigm for applying intelligent methods to solve real-time scheduling problems of pump and valve.

2. METHODOLOGY

2.1. Leakage modelling

A hydraulic simulation, including background leakage, is essential for evaluating the water loss element of the objective. Background leakage is caused by small outflows from cracks and fittings. [Germanopoulos \(1985\)](#) proposed a model to compute background leakages at the pipe level as follows:

$$d_k^{\text{leaks}}(P_{k,\text{mean}}) = \begin{cases} \beta_k L_k P_{k,\text{mean}}^{\alpha_k} & P_{k,\text{mean}} > 0 \\ 0 & P_{k,\text{mean}} < 0 \end{cases} \quad (1)$$

where k is the index of the pipe, d_k^{leaks} is the background leakages along the k th pipe, $P_{k,\text{mean}}$ is the mean pressure along the k th pipe, L_k is the length of the k th pipe, and α_k and β_k are the model parameters. The exponent α_k depends on the pipe material and rigidity, and the value of α_k has been reported ranging from 0.50 to as high as 2.50, depending on the mixture of leaks and the dominant type of leaks ([Lambert 2001](#)). Plastic pipes generally have higher α_k values and the rigid pipes have lower α_k values ([Giustolisi et al. 2008](#)). In the case study, a value of $\alpha_k = 1.2$ was selected according to [Giustolisi et al. \(2013\)](#), [Jowitt & Xu \(1990\)](#) and [Vairavamoorthy & Lumbers \(1998\)](#). The parameter β_k depends not only on the pipe characteristics but also on the asset condition. The value of β_k is usually assumed close to zero in systems where water utilities routinely conduct leakage management ([Giustolisi et al. 2008](#)). In the case study, to test different levels of leakages, the β_k values were obtained as $\beta_k = 7.38 \times 10^{-9}$ and $\beta_k = 1.67 \times 10^{-8}$ by forcing the total leakage flow rate to 10 and 20% of the total demand.

The mean pressure of the pipe is given by the pressure of the end nodes i and j of the k th pipe as:

$$d_k^{\text{leaks}}(P_{k,\text{mean}}) = \beta_k L_k \left(\frac{P_i + P_j}{2} \right)^{\alpha_k} \quad (2)$$

For modelling purposes, the background leakage outflow is concentrated at the two end nodes and equally divided as:

$$d_i^{\text{leaks}}(P_{k,\text{mean}}) = d_j^{\text{leaks}}(P_{k,\text{mean}}) = \frac{\beta_k L_k}{2} \left(\frac{P_i + P_j}{2} \right)^{\alpha_k} \quad (3)$$

EPANET was used as a hydraulic solver in this study; however, EPANET does not support pressure-dependent pipe leakage in the mass balance equations. Thus, an equivalent method using an emitter was used to simulate the background leakage and allocate the pressure-dependent leakage demand to the node. The leakage at a particular node is a function of the

pressure at a single node:

$$d_i^{\text{leaks}'}(P_i) = \frac{\beta_k L_k}{2} (P_i)^{\alpha_k} \quad (4)$$

$$d_j^{\text{leaks}'}(P_j) = \frac{\beta_k L_k}{2} (P_j)^{\alpha_k} \quad (5)$$

The total leakage of the k th pipe is:

$$d_k^{\text{leaks}'}(P_i, P_j) = \beta_k L_k \frac{(P_i)^{\alpha_k} + (P_j)^{\alpha_k}}{2} \quad (6)$$

This modelling process generates an error between $d_k^{\text{leaks}'}(P_i, P_j)$ and $d_k^{\text{leaks}}(P_{k,\text{mean}})$, which may influence the accuracy of estimation of water loss (Giustolisi *et al.* 2016). Therefore, this study first verified the accuracy of the emitter-based leakage model.

The error of the emitter-based leakage model can be expressed as:

$$E_{d_k^{\text{leaks}}} = \left| \frac{d_k^{\text{leaks}'}(P_i, P_j) - d_k^{\text{leaks}}(P_{k,\text{mean}})}{d_k^{\text{leaks}}(P_{k,\text{mean}})} \right| = \left| \frac{d_k^{\text{leaks}'}(P_i, P_j)}{d_k^{\text{leaks}}(P_{k,\text{mean}})} - 1 \right| \quad (7)$$

where

$$\begin{aligned} \frac{d_k^{\text{leaks}'}(P_i, P_j)}{d_k^{\text{leaks}}(P_{k,\text{mean}})} &= \frac{\beta_k L_k \frac{(P_i)^{\alpha_k} + (P_j)^{\alpha_k}}{2}}{\beta_k L_k \left(\frac{P_i + P_j}{2} \right)^{\alpha_k}} \\ &= 2^{\alpha_k - 1} \frac{1 + \left(\frac{P_j}{P_i} \right)^{\alpha_k}}{\left(1 + \frac{P_j}{P_i} \right)^{\alpha_k}} \end{aligned} \quad (8)$$

There are head losses along the pipes, and the pressures at end nodes of a pipe are not equal. Assuming that $P_j > P_i$ and $x = P_j/P_i$, it can be inferred that $x > 1$.

Let

$$g(x) = \frac{1 + x^{\alpha_k}}{(1 + x)^{\alpha_k}} \quad (9)$$

By taking the derivative of $g(x)$, we obtain

$$g'(x) = \frac{\alpha_k (1 + x)^{\alpha_k - 1} (x^{\alpha_k - 1} - 1)}{(1 + x)^{2\alpha_k}} \quad (10)$$

Since $\alpha_k = 1.2$ in this study, when $x > 1$, $g'(x) > 0$, function $g(x)$ is monotonically increasing, and $g(x) > g(1) = 1$. The error in the alternative emitter model can be expressed as:

$$E_{d_k^{\text{leaks}}} = 2^{\alpha_k - 1} g(x) - 1 \quad (11)$$

The larger the P_j/P_i , the larger will be $E_{d_k^{\text{leaks}}}$. If $P_j/P_i = 2$, the error $E_{d_k^{\text{leaks}}} = 1.35\%$, and the error $E_{d_k^{\text{leaks}}} = 3.10\%$ if $P_j/P_i = 3$. In this study, the objective problem has a minimum pressure constraint of 14 m. After the pre-test, the pressure P_j was no more than three times of P_i , except for pressure of the end nodes of connection pipes of the pumps and tanks in the case

network. Therefore, we modelled the background leakage by adding two emitters to the end nodes of the pipe, which was verified to have sufficient accuracy, with an error below 3.10%. We also computed the error of different α_k , as shown in Supplementary Figure S1. Based on Figure S1, the error significantly increases when $\alpha_k \geq 1.5$. It is not recommended to use the emitter-based leakage model when $\alpha_k \geq 1.5$, which will greatly affect the results of leakage computation in the hydraulic model of WDNs.

2.2. Problem formulation

To achieve water resource management of background leakages and energy management of pumps considering the hydraulic requirements, the objective function of the optimisation problem of the pump and valve scheduling minimises the two main objectives: energy cost and water loss cost, as shown in the following equation:

$$\min f(\text{Energy cost, water losses cost}) = \min \sum_{t=1}^{t=24} \sum_{p=1}^{P=p_{\text{pump}}} \frac{\rho g H_{p,t} Q_{p,t}}{\eta} C_e \Delta T + C_l \sum_{t=1}^{t=24} \sum_{i=1}^{i=I} \frac{\beta_k L_i}{2} P_{i,t}^{\alpha_k} \Delta T \quad (12)$$

The energy cost of the pump depends on its power. In Equation (12), ρ is the density of water, g is the gravity constant, $H_{p,t}$ and $Q_{p,t}$ are the head and flow, respectively, of p th pump at time t , η is the efficiency, C_e is the energy tariff, and ΔT is the duration. C_l is the water loss cost per cubic metre.

The model constraints are shown in Equations (13)–(17). Equations (13) and (14) represent the continuities of mass and energy, respectively. Equation (15) expresses the minimum and maximum constraints at the tank level. These three requirements are incorporated into the EPANET solver. Equation (16) represents the pressure requirement. Equation (17) limits the range of the final tank level, which should be higher than the initial level, to exclude the solutions of water overconsumption in the tanks:

$$\sum_{j \in J(i)} q_{ij} = d_i + d_i^{\text{leaks}} \quad (13)$$

$$\pi_{ij} = h_i - h_j \quad (14)$$

$$H_{\text{tank,min}} \leq H_{\text{tank},t} \leq H_{\text{tank,max}} \quad (15)$$

$$H_{\text{min}} \leq H_i \leq H_{\text{max}} \quad (16)$$

$$H_{\text{tank,fin}} \geq H_{\text{tank,ini}} \quad (17)$$

where d_i is the demand at node i , d_i^{leaks} is the background leakage, q_{ij} is the algebraic flow to node i , $J(i)$ is the set of adjacent node to node i , π_{ij} is the head loss in the pipe connecting nodes i and j , and H_{tank} , $H_{\text{tank,min}}$, and $H_{\text{tank,max}}$ are the current tank level at time t , minimum tank level, and maximum tank level, respectively. $H_{\text{tank,ini}}$ is the initial tank level and $H_{\text{tank,fin}}$ is the final tank level. H_i is the pressure of node i , and H_{min} and H_{max} are the minimum and maximum pressure requirements, respectively, of the WDN.

The decision variables for the optimisation problem of the pump and valve scheduling are the status of the pumps and control valve settings of the PRVs in the WDN. The status of the pumps not only affects its energy consumption but also influences the pressure and water losses level in the WDN. Valve operations, through their impact on the pressure at nodes downstream of the valves, affect the water losses cost. The pumps and valves work together to achieve the goals of water resource and energy management.

2.3. Markov decision process

Modelling a Markov decision process (MDP) (Bäuerle & Rieder 2010; Puterman 2014) is an essential step for solving optimisation problems using multi-agent methods. In this problem, we considered two types of agents designated as $y \in \{\text{pump, valve}\}$. The agents learnt their policies through interactions with the environment. At each time step, each agent observed the current state of the environment and selected the action following the policy. The environment was affected by the joint action and transitioned to a new state, rewarding each agent. Then the agents moved to the next state. During the interaction process, the agents adjusted their policies based on the feedback of the environment. MDP comprises three critical components: state, reward, and action. These components were specified as follows.

2.3.1. State

In contrast to traditional optimisation methods, the multi-agent reinforcement learning method can perceive changes in the environment and find real-time optimal solutions which can adapt to these changes. The state includes all the information relevant to the environment perceived by the agent.

In this study, the water demand s_{demand} and tank level s_{tank} were applied as the environmental states for the cooperative decision-making of pumps and valves. We followed a three-step approach to consider time and space uncertainties in water demand and construct an environment with an uncertain water demand state.

First, the water demand pattern $p'_{at,t}$ with time uncertainty was constructed:

$$p'_{at,t} = p_{at,t} \times \varepsilon_t^{\text{time}} \quad (18)$$

where $p_{at,t}$ is the original demand pattern at time step t and $\varepsilon_t^{\text{time}}$ is the time uncertainty multiplier.

Second, the base demand $d_i^{\text{base}'}$ with spatial uncertainty was constructed:

$$d_i^{\text{base}'} = d_i^{\text{base}} \times \varepsilon_i^{\text{space}} \quad (19)$$

where d_i^{base} is the original base demand of the i th node and $\varepsilon_i^{\text{space}}$ is the space uncertainty multiplier of node i .

Third, the water demand $d'_{i,t}$ of the i th node at time step t was calculated as:

$$d'_{i,t} = p'_{at,t} \times d_i^{\text{base}'} \quad (20)$$

The uncertainty multipliers $\varepsilon_t^{\text{time}}$, $\varepsilon_i^{\text{space}}$ follow a truncated normal distribution with a range from 0.7 to 1.3. The water demand states were normalised between zero and one for each consumer node. As the initial tank level varies daily, we assume that the initial tank level of the environment follows a uniform distribution to account for this fluctuation, where $s_{\text{tank},t=0} \in [0, 1]$. The subsequent tank level is determined by the previous tank-level state and the joint action taken by all agents. The final state space setting is:

$$S = (s_{\text{demand}1}, s_{\text{demand}2}, \dots, s_{\text{demand}N}, s_{\text{tank}1}, s_{\text{tank}2}, \dots, s_{\text{tank}Z}) \quad (21)$$

2.3.2. Reward

A reward is the feedback that agents receive from the environment after taking action; it reflects the quality of the action. Sparse rewards can result in a slow learning process. It is necessary to design a reward at each time step to provide timely feedback to the agents based on the objective function of the task. In this scheduling problem, the lower the cost, the better is the scheduling solution, which is contrary to the concept of a multi-agent reward. Therefore, agents consider random interactions with the environment to obtain a benchmark cost. Then, this benchmark is subtracted by the cost generated at each time step to construct the reward. In addition, penalties are imposed on actions that do not satisfy the hydraulic and tank-level constraints.

(1) Reward for action when hydraulic constraints are satisfied:

$$r_{m,t} = \begin{cases} \frac{r_{\text{benchmark}}}{24} - \sum_{p=1}^{P=P_{\text{pump}}} \frac{\rho g H_{p,t} Q_{p,t}}{\eta} C_e \Delta T - C_l \sum_{i=1}^{i=I} \frac{\beta_k L_i}{2} P_{i,t}^{\alpha_k} \Delta T & t < 24 \\ \frac{r_{\text{benchmark}}}{24} - \sum_{p=1}^{P=P_{\text{pump}}} \frac{\rho g H_{p,t} Q_{p,t}}{\eta} C_e \Delta T - C_l \sum_{i=1}^{i=I} \frac{\beta_k L_i}{2} P_{i,t}^{\alpha_k} \Delta T + C_{\text{tank}} & t = 24 \end{cases} \quad (22)$$

$$C_{\text{tank}} = \begin{cases} \vartheta \frac{(V_{\text{fin}} - V_{\text{ini}})}{V_{\text{ini}}} r_{\text{benchmark}} & V_{\text{fin}} < V_{\text{ini}} \\ 0 & V_{\text{fin}} \geq V_{\text{ini}} \end{cases} \quad (23)$$

where $r_{\text{benchmark}}$ is the reward benchmark that converts the cost into a reward form, $r_{m,t}$ is the reward of the m th agent at time step t , and C_{tank} is the penalty for the reduction in the final tank level V_{fin} compared with the initial tank level V_{ini} .

(2) Reward for action when hydraulic constraints are violated:

$$r_{m,t} = C_{\text{hydraulic}} \quad (24)$$

where $C_{\text{hydraulic}}$ is the penalty for the network hydraulic constraints. The settings for $r_{\text{benchmark}}$, $C_{\text{hydraulic}}$, and ϑ are listed in Supplementary Table S1.

2.3.3. Action

An action space is a set of actions that can be performed by all agents. In optimisation scheduling, the decision variables are mapped to the actions of the agents using the multi-agent reinforcement learning method. Specifically, we treated pumps and valves as heterogeneous agents owing to their distinct characteristics. The pump was modelled as a variable-speed pump with a speed range of 0.7–1 considering mechanical constraints. The action of the pressure reducing valve (PRV) agent was to set the outlet pressure, and its action space was determined based on the minimum pressure requirement of consumers and the valve inlet pressure. The valve inlet pressure of the PRV was calculated using 1,000 random interaction episodes with the environment, and the maximum valve inlet pressure was set as the upper limit of the action space of the PRV agent, as listed in Supplementary Table S2. In this study, the minimum pressure requirement for the consumers was 14 m.

2.4. MADDPG

The optimisation of the pump and valve operation is different from that of a single-agent system. The dependencies across different agents may affect flow allocations and pressure distributions. Single reinforcement learning may encounter problems regarding the dimensionality and the dynamic environment for multi-agents. The multi-agent framework can deal with the cooperation of pumps and valves as well as the uncertainties in the environment. The water demand and tank-level fluctuate during daily scheduling, and an appropriate scheduling policy for the WDN needs to be specifically tailored to the unique conditions of the water demand and tank level. Time plays a crucial role in real-time scheduling. Traditional optimisation methods cannot adapt to the changing of the environment state timely. The multi-agent framework has the potential for generalisation and is well suited for adapting to the uncertainty of the environment by interacting with the uncertain WDN environment. After the training process, each agent will obtain a policy that can give suitable solutions for new problem instances under the uncertain environment within a short time.

We designed an MADDPG framework, which is an extension of the actor-critic method (Lowe *et al.* 2017). The critic network of each agent obtains observation and action information from all agents to eliminate the non-stationarity of the multi-agent environment. The parameters of the centralised critic network are updated by minimising the loss:

$$L_{\theta_m^Q} = \frac{1}{N} \sum_{n=1}^N (y^n - Q_m^\mu(s^n, a_1^n, a_2^n, \dots, a_M^n | \theta_m^Q))^2 \quad (25)$$

where s is the global observation of all agents, $a_1, a_2, \dots, a_M$ is the joint action, $Q_m^\mu(s^n, a_1^n, a_2^n, \dots, a_M^n | \theta_m^Q)$ is the joint state-action value calculated by the critic network of the m th agent, and y^n is the target value obtained from the Bellman equation, which is expressed as:

$$y^n = r_m^n + \gamma Q_m^{\mu'}(s'^n, a_1'^n, a_2'^n, \dots, a_M'^n | \theta_m^Q) \Big|_{a'_m = \mu'_m(s'_m | \theta_m^{\mu'})} \quad (26)$$

where $Q_m^{\mu'}(s'^n, a_1'^n, a_2'^n, \dots, a_M'^n | \theta_m^Q) \Big|_{a'_m = \mu'_m(s'_m | \theta_m^{\mu'})}$ is the joint state-action target value of the next state, r_m^n is the reward of the m th agent, and γ is the discount factor.

The actor network of each agent gets the observation on its own and calculates the action by the policy $\mu_m(s_m | \theta_m^\mu)$. The parameters of the actor network are updated in the direction of increasing value using the gradient descent method:

$$\nabla_{\theta_m^\mu} J = \frac{1}{N} \sum_{n=1}^N \nabla_{a_m} Q_m^\mu(s^n, a_1^n, a_2^n, \dots, a_M^n | \theta_m^Q) \Big|_{a_m = \mu_m(s_m | \theta_m^\mu)} \cdot \nabla_{\theta_m^\mu} \mu_m(s_m | \theta_m^\mu). \quad (27)$$

The parameters of target networks $\theta_m^{Q'}$ and $\theta_m^{\mu'}$ are updated based on the parameters of evaluated networks using the soft update method:

$$\theta_m^{Q'} \leftarrow \tau \theta_m^Q + (1 - \tau) \theta_m^{Q'}, \quad (28)$$

$$\theta_m^{\mu'} \leftarrow \tau \theta_m^{\mu} + (1 - \tau) \theta_m^{\mu'}, \quad (29)$$

where $\tau \ll 1$.

The training procedure of MADDPG is shown in Algorithm 1.

Algorithm 1. Training procedure of MADDPG

- 1: Initialisation: randomly initialise weights for critic network θ_m^Q and actor network θ_m^{μ} , and initialise target critic network as $\theta_m^{Q'} = \theta_m^Q$ and target actor network as $\theta_m^{\mu'} = \theta_m^{\mu}$. Initialise experience replay buffer.
 - 2: **for** episode = 1 to E **do**
 - 3: Initialise a random process for multi-agent energy and water resources management action exploration and receive the initial state s .
 - 4: **for** step $t = 1$ to 24 **do**
 - 5: For each agent m , receive the observation and output action $a_m = \mu_m(s_m | \theta_m^{\mu}) + \mathcal{N}_t$
 - 6: Execute joint action $a = (a_1, a_2, \dots, a_M)$ and collect reward r_m and next state s'_m for each agent.
 - 7: Store the transition tuple (s, a, r, s') in the experience replay buffer.
 - 8: Update state $s \leftarrow s'$.
 - 9: **for** agent $m = 1$ to M **do**
 - 10: Sample a minibatch of transition tuple (s^n, a^n, r^n, s'^n) from experience replay buffer.
 - 11: Update critic network using Equations (25) and (26)
 - 12: Update actor network using Equation (27)
 - 13: **end for**
 - 14: Update the target critic and actor parameter for each agent m using Equations (28) and (29)
 - 15: **end for**
 - 16: **end for**
-

The number of episodes in the training procedure of MADDPG was set to 20,000. The MADDPG hyperparameters were fine-tuned and are listed in Supplementary Table S3.

In the application process, each agent receives the observation from the environment and independently selects an action according to a trained actor network, without relying on the centralised critic network. The trained model obtained real-time scheduling on the second scale. This study applied a fully cooperative multi-agent task. Each agent receives the same reward and state and shares the critic network, significantly improving the training efficiency. However, the agents used were not homogeneous. Each agent had a different effect on the critic network, affecting its actor network.

3. RESULTS AND DISCUSSION

3.1. Introducing the studied WDN

EPANET Net 3 was trained and tested in this study. The EPANET Net 3 network is one of the most popular benchmarks for WDN optimisation tasks because of its data availability and flexibility for modification (Mala-Jetmarova *et al.* 2017). The numerical model is accessible online from the web page of the Kentucky Hydraulic Model Database (Ormsbee *et al.* 2022). The network has two raw water sources, two pump stations, and three tanks. Figure 1 shows the topology of EPANET Net 3. To apply the proposed method, the network was modified as follows:

- (1) Four valves with the greatest impact on leakage were added to the network through a pre-experiment using a GA.

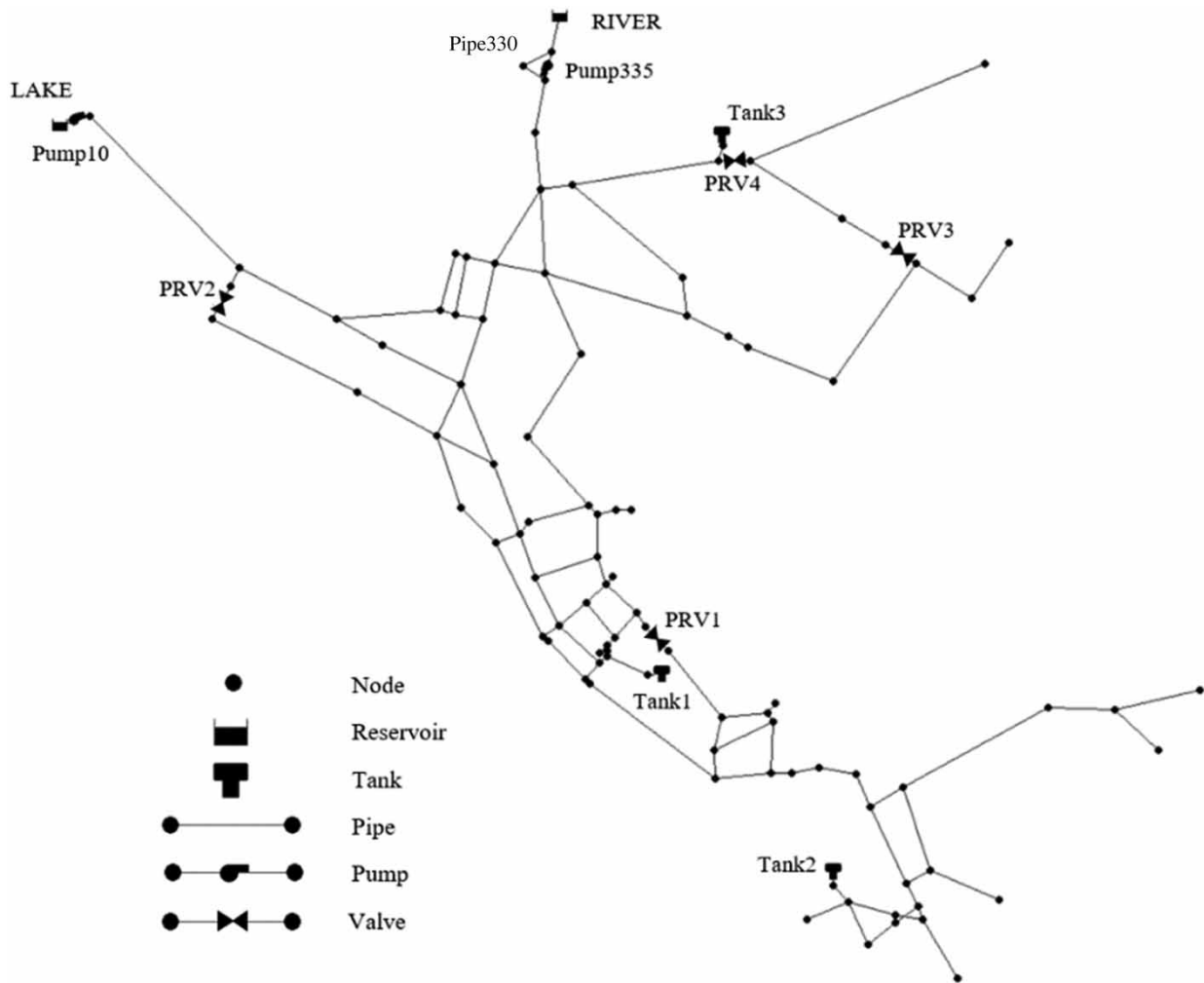


Figure 1 | Topology of EPANET Net 3.

- (2) All simple control rules of pumps 10, 335, and pipe 330 were deleted. Pipe 330 was closed; this meant that the lake and river supplied water only through pumps 10 and 335, respectively. Pumps 10 and 335 were not controlled by the rules but by multi-pump agents.
- (3) A background leakage was added to every node downstream of the pumps as an emitter, as described in Section 2.1.
- (4) Demand variations were added, as shown in Section 2.3.1.
- (5) The general default demand pattern with a single peak was modified to a double-peak pattern, as shown in Figure 2.

In this study, time from 7:00 to 23:00 h was on-peak with a peak tariff of 0.1194 \$/kWh, while 0:00 to 7:00 h and 23:00 to 24:00 h were off-peak with an off-peak tariff of 0.0244 \$/kWh (Bagirov *et al.* 2013). As background leakage is an important component of the non-revenue water, the value of a m³ of background water loss cost was assumed to be 0.31 \$/m³ according to Liemberger & Wyatt (2018). The wire-to-water efficiency of the pumps was simplified to 0.75. All the test cases are randomly generated from the same distribution as the training cases. Specifically, the uncertainty multipliers ϵ_i^{time} and $\epsilon_i^{\text{space}}$ are randomly chosen between 0.7 and 1.3, following a truncated normal distribution. The initial tank level is randomly chosen between 0 and 1, following a uniform distribution.

All the models were trained using a computer with an Intel Core i7-11700K central CPU. The MADDPG models were coded with Pytorch, and the WDN environment was built using Water Network Tool for Resilience (WNTR), which is

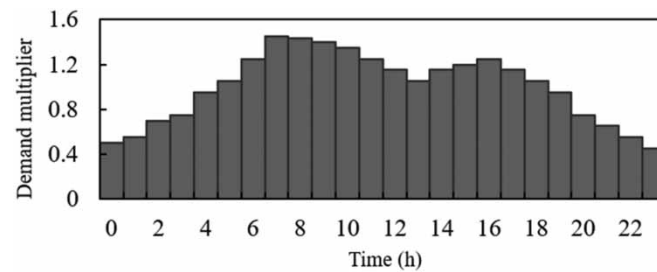


Figure 2 | The general default demand pattern of EPANET Net 3.

compatible with EPANET under Python version 3.9. The metaheuristic models were built using MEALPY version 2.5.1, a Python library of metaheuristic algorithms, including GA, PSO, and DE.

3.2. Effects of energy and water loss costs

We conducted 100 random test cases from the same distribution of the training cases under different leakage levels to study the relationship between the objective of the energy cost and water loss cost. Figures 3 and 4 show the optimal pattern of pump scheduling, water losses, and variation ratio of the tank level over time in the case of water loss levels of 10 and 20%, respectively. Each figure is composed of nine diagrams: the top three are the results of minimising the energy cost and water losses through the pumps and valves; the middle three are the results achieved only by pumps; and the bottom three are the results achieved by using only pumps and neglecting the water loss objective.

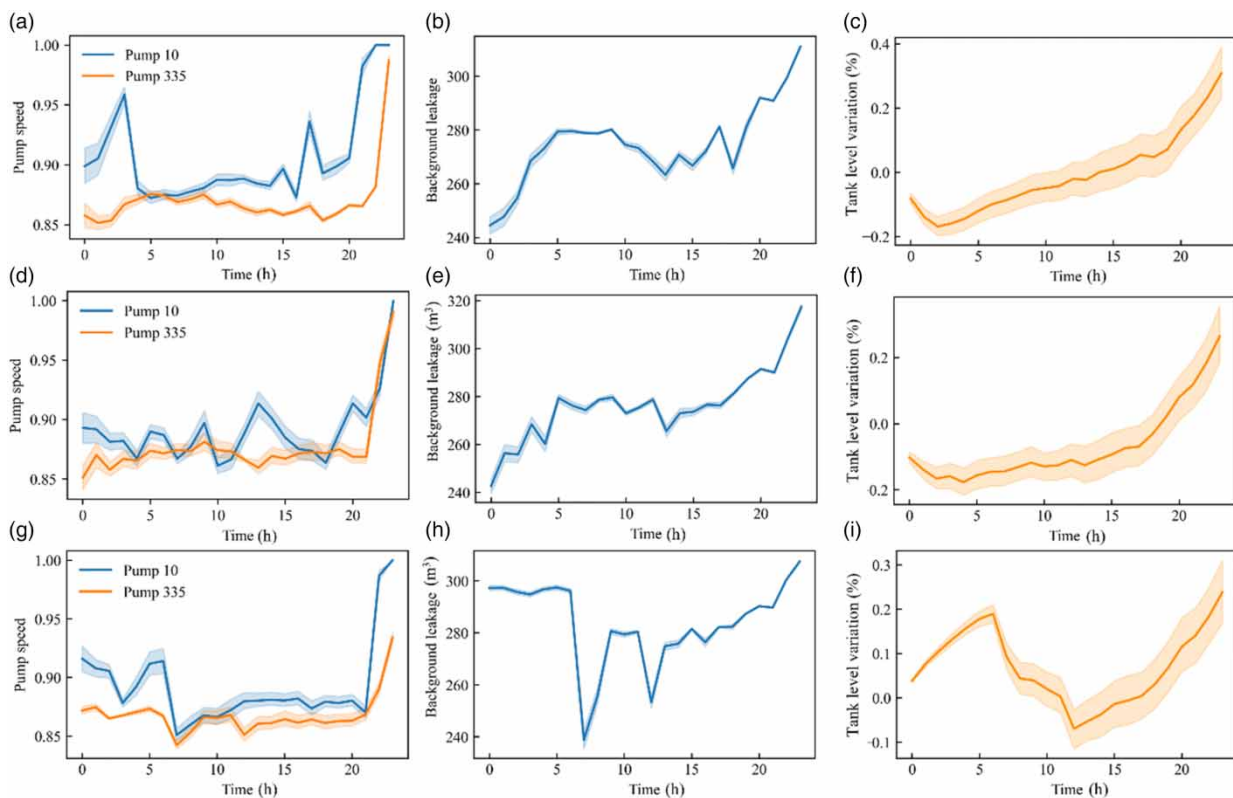


Figure 3 | Pump speed, background leakage, and tank-level variation ratio over time for water losses of approximately 10%: (a)–(c) Minimising the energy and water loss costs using the pump and valve (d)–(f) Minimising the energy and water loss costs using the pump (g)–(i) Minimising the energy cost using the pump and neglecting the water loss objective.

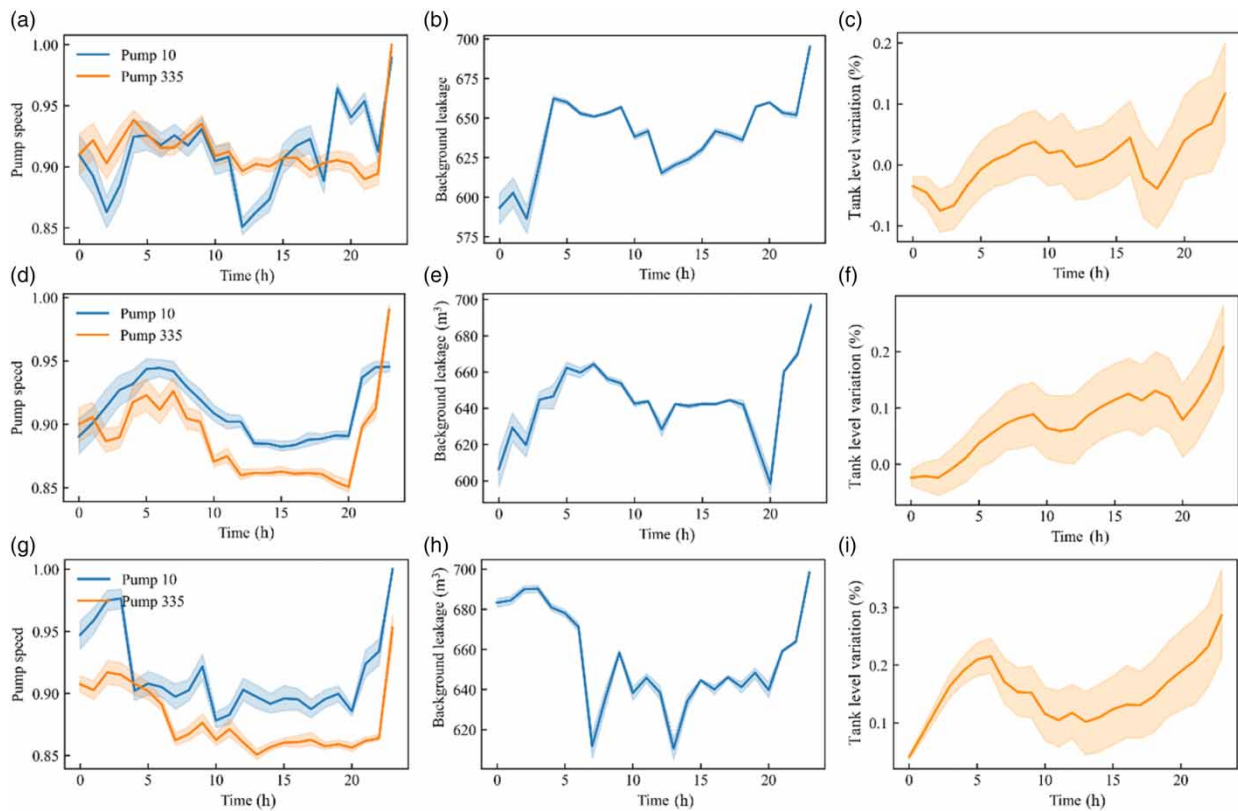


Figure 4 | Pump speed, background leakage, and tank-level variation ratio over time for water losses of approximately 20%: (a)–(c) Minimising the energy and water loss costs using the pump and valve (d)–(f) Minimising the energy and water loss costs using the pump (g)–(i) Minimising the energy cost using the pump and neglecting the water loss objective.

Table 1 lists the detailed information of the results for the three random cases from the same distribution of the training cases for the different optimisation tasks. The labels E + W + V and E + W in the third column indicate the results achieved by minimising the energy and water loss costs through pump and valve scheduling and pump scheduling, respectively. Label E is the result of minimising the energy cost through pump scheduling alone.

Figures 3(g)–3(i) and 4(g)–4(i) show that the scheduling patterns for minimising the energy cost by optimal pump scheduling are significantly affected by the energy tariff. Water losses during off-peak electricity hours are much higher than those during peak electricity hours, which are caused by pumping during off-peak electricity hours to store water. During 0:00–7:00 h, the pumps filled the tank to reach a peak level at 7:00 h. Then, the water in the tank was used to assist the pumping systems during peak demand hours from approximately 7:00 to 12:00 h. During the second peak hours, the average pressure level was similar to that during the first peak hours because the water losses did not differ significantly. The pumps provide sufficient water to satisfy the demand during the second peak hours without supplement from the tank. The tank is filled from 13:00 h to reach the initial level, especially 23:00–24:00 h, the pump speed had an apparent increase in storing the water in the tank during off-peak electricity hours.

In contrast, the scheduling patterns of minimising the energy and water loss costs through pump and valve scheduling (Figures 3(a)–3(c) and 4(a)–4(c)) or only pump scheduling (Figures 3(d)–3(f) and 4(d)–4(f)) are less affected by the electricity tariff but are mainly affected by the demand pattern of the consumers. During 0:00–16:00 h, the variation tendency of water losses and tank levels for a water loss level of 20% is similar to the demand pattern, indicating that the pump pattern followed the demand pattern. Rather than pumping at night, the optimal schedule was to pump during peak demand hours to minimise the total cost. For a water loss level of 10%, the water in the tank was used from 0:00 to 3:00 h (off-peak electricity hours) and filled in the following hours. In all cases, the pump speed had an apparent increase during 23:00–24:00 h because the agent learnt to store as much water as possible at the last time step of the day to avoid tank-level punishment.

Table 1 | Energy cost, water loss cost, and total cost for different optimisation tasks

Parameter of leakage model	Case	Minima	Energy cost (\$/day)	Water losses cost (\$/day)	Total cost (\$/day)
$\beta_k = 7.38 \times 10^{-9}$	1	E + W + V	254.74	2,022.68	2,277.42
		E + W	252.85	2,038.91	2,291.76
		E	199.47	2,116.56	2,316.03
	2	E + W + V	241.87	1,999.73	2,241.60
		E + W	261.21	2,011.69	2,272.90
		E	201.27	2,116.99	2,318.26
	3	E + W + V	286.43	2,070.43	2,356.86
		E + W	274.55	2,086.22	2,360.77
		E	251.76	2,115.20	2,366.96
$\beta_k = 1.67 \times 10^{-8}$	1	E + W + V	337.67	4,653.54	4,991.20
		E + W	286.43	4,745.84	5,032.26
		E	242.80	4,893.63	5,136.44
	2	E + W + V	385.46	4,656.53	5,041.98
		E + W	327.80	4,721.99	5,049.79
		E	261.81	4,865.01	5,126.81
	3	E + W + V	358.35	4,762.54	5,120.90
		E + W	298.52	4,848.66	5,147.18
		E	276.51	4,881.67	5,158.19

Table 1 lists the costs of the different scheduling tasks. The total cost of minimising the energy cost using the optimal pump schedule was greater than that of the optimisation tasks, which also accounted for the water loss cost. Although pumping during off-peak hours reduced the energy cost, it increased the water loss, increasing the total cost. The scheduling patterns for minimising energy and water loss costs followed the law of the demand pattern and achieved a lower total cost. The reduction in total cost was more evident in the case of a water loss level of 20%. The results indicate that the water loss cost dominates the energy cost in minimising the total cost of scheduling tasks, although the unit volume cost of non-revenue water is assumed to be low. The energy cost of minimising the energy and water loss costs through the pump and valve scheduling task is the highest among the different optimisation tasks. With the cooperation of the pump and valve, the cost of water losses can be greatly reduced to obtain the lowest total cost of 2,291.96 and 5,051.36 \$/day on average for water loss levels of 10 and 20%, respectively.

3.3. Comparison of models

To test the performance of the multi-agent reinforcement, metaheuristic models were used as benchmark models, as they have been proven to be efficient in WDN scheduling problems and are mostly commonly used in the field (Mala-Jetmarova *et al.* 2017). We tested the MADDPG, GA, PSO, and DE models for five random test cases from the same distribution of the training cases under different leakage levels. The results are presented in Tables 2 and 3. Table 2 lists the total costs, including energy and water loss costs, and Table 3 lists the average training time and average solving time for test cases of the different models. The MADDPG model has the ability for generalisation. The training and application processes of the MADDPG

Table 2 | Total cost of different models (\$/day)

Parameter of leakage model	Model	Case1	Case2	Case3	Case4	Case5
$\beta_k = 7.38 \times 10^{-9}$	MADDPG	2,241.60	2,307.22	2,277.42	2,281.93	2,323.39
	GA	2,377.84	2,289.65	2,324.11	2,465.91	2,301.72
	PSO	2,607.41	2,505.46	2,541.68	2,527.69	2,507.55
	DE	2,587.59	2,513.24	2,569.43	2,465.91	2,522.21
$\beta_k = 1.67 \times 10^{-8}$	MADDPG	5,011.37	4,991.20	5,041.98	5,096.60	5,106.37
	GA	4,962.96	5,108.92	5,006.78	5,101.74	5,075.21
	PSO	5,290.90	5,393.60	5,311.20	5,393.63	5,233.07
	DE	5,250.83	5,371.89	5,349.33	5,407.22	5,296.37

Table 3 | Computation time of different models (s)

Model	Average training time	Average solving time for test cases
MADDPG	46,868.35	1.45
GA	–	8,030.04
PSO	–	669.02
DE	–	537.47

model were separate. MADDPG trains agents offline. Once the MADDPG model is trained, agents can adapt to different test cases of the uncertain environment and find optimal solutions within a short time. However, traditional metaheuristic methods cannot adapt to dynamic environments; therefore, these methods must solve an optimisation schedule for each test case. The epoch and population sizes were 500 and 50, respectively, for the GA, PSO, and DE models, and early stopping was applied to reduce the computation time. If the global best did not improve after 30 epochs, the program was terminated. For GA, the cross-over probability and mutation probability was 0.95 and 0.025, respectively. The weight min of the bird and weight max of the bird of PSO was 0.4 and 0.9, respectively. For DE, the cross-over rate was 0.9 and the weighting factor was 0.8.

From Table 2, among the models, MADDPG is the best, followed by GA, PSO, and DE. At a water loss level of 10%, the average total costs of the MADDPG and GA were 2,286 and 2,352 \$/day, respectively. For a water loss level of 20%, the average total costs of MADDPG and GA were 5,049 and 5,051 \$/day, respectively. The PSO and DE models were prone to local optimal solutions with limited parameter tuning. When the training time was about 600 s, which is much lower than the 8,030 s of the GA, the convergence condition of the early stop of the PSO and DE models was achieved, and the total costs were much higher than that of the GA and MADDPG.

As shown in Table 3, computing for approximately 8,030.04 s is necessary for GA models to obtain the optimal solution for each case, which is impractical for daily scheduling at hourly intervals. Though the MADDPG models need a lot of training time, it can adapt with the uncertainty of the water demand and the initial tank level in the WDN. In the application process, the MADDPG models can provide results comparable to the GA models in only 1.45 s, which shows the great advantage of the MADDPG. MADDPG can help build an intelligent scheduling system and make optimal real-time decisions.

4. CONCLUSIONS

This study proposed a multi-agent reinforcement learning framework for real-time scheduling of pump and valve. The objective was to minimise the energy and water loss costs using the pump and valve together with the help of collaborative intelligent scheduling agents. The scheduling problem was modelled as an MDP to apply the multi-agent reinforcement learning method. Then, we explored the impact of water loss and energy costs on the scheduling strategy. The results of the proposed MADDPG method were compared with those of traditional methods, such as GA, DE, and PSO. The results show that the scheduling solutions for minimising water loss and energy costs are affected by the demand pattern of consumers rather than the energy tariff. The water loss cost dominates the energy cost in minimising the total cost of the scheduling tasks. MADDPG can flexibly deal with the uncertainty of water demand and obtain the optimal solution. Although GA models have comparable performance to MADDPG, it took much more time to obtain the optimal solution for each case. The multi-agent reinforcement learning framework has broad application prospects in complex real-time scheduling problems of WDNs.

In this study, we applied a simplified leakage model by adding two emitters to the end nodes of the pipe. The calibration part of the hydraulic model will be further considered in future works to accurately simulate the leakage of the WDN. The tariff changes for different seasons or the network topology changes are not considered in this study, the MADDPG model should be retrained when these changes occur. These uncertainties will also be considered in future studies.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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